
Improving Context Grounding of LLM Text Generation

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Abstract

Retrieval-augmented generation requires language models to prioritize retrieved context over memorized knowledge — yet models frequently fail to do so, and no retraining-free mechanism for correcting this at inference time exists. We show that context-grounding behavior is linearly encoded in attention-head activations, and that this encoding can be directly exploited: injecting probe-derived directions as permanent activation biases increases context adherence by up to 5.8 percentage points on knowledge-conflict inputs, with no regression on factually consistent ones, though intervention strength requires careful tuning to avoid catastrophic generation failure. Crucially, the mechanism is *unconditional* — it amplifies adherence to whatever context is provided regardless of factual correctness — explaining both the gain on conflicting inputs and the absence of improvement on consistent ones. Probe analysis further reveals that context-grounding representations subsume truthfulness representations: virtually no attention head is dedicated exclusively to truthfulness, suggesting both properties can be steered through the same internal components. Finally, the trained probes serve as a single-pass attribution tool for RAG applications, identifying per token whether a model’s response relies on retrieved context or parametric knowledge.

1 Introduction

Large language models (LLMs) are widely deployed in retrieval-augmented generation (RAG) systems [Shuster et al., 2021], where relevant documents are prepended to the model’s input at query time. This allows access to up-to-date or domain-specific information without retraining. However, providing context does not guarantee that the model will use it. LLMs encode factual associations during pretraining — referred to as *parametric knowledge* — and this internal knowledge can override information supplied at inference time [Xie et al., 2024, Ju et al., 2024]. When context contradicts memorized beliefs, models may disregard it entirely or produce responses that seem plausible but are unfaithful to the source. In high-stakes settings such as legal document analysis or medical question answering, such failures are directly harmful.

Prior work has shown that factual knowledge and contextual information are predominantly encoded in the higher layers of transformer models [Meng et al., 2022, Dai et al., 2022, Ju et al., 2024]. Building on this, Li et al. [2023] introduce *inference-time intervention* (ITI). Linear probes are trained on attention-head activations to identify heads associated with truthful generation, and a scaled probe direction is injected into those activations at inference time, steering outputs toward more truthful responses. On TruthfulQA [Lin et al., 2022], ITI improves LLaMA-7B truthfulness from 32.5% to 65.1%; however, this large margin partly reflects the relatively low baseline of that earlier model. A related decoding-time method, DoLa [Chuang et al., 2024], contrasts higher and lower layer outputs to promote factual tokens, improving TruthfulQA performance by 12–17%. Despite these advances, ITI has only been evaluated for factual truthfulness. Whether the same mechanism can improve

38 *context-grounded* generation — where the model must follow provided context rather than internal
39 knowledge — remains unexplored.

40 This paper extends ITI to context-grounded generation. We train linear probes on LLaMA-3-
41 8B-Instruct [Grattafiori et al., 2024] activations using ConflictQA [Xie et al., 2024], a dataset
42 of knowledge-conflict question-answer pairs, and apply interventions at inference time. Experi-
43 ments show that the improvement is concentrated on inputs where context conflicts with parametric
44 knowledge: false-context samples improve by 5.8 percentage points while true-context samples
45 are unaffected, revealing that the intervention amplifies context adherence unconditionally rather
46 than selectively. We further compare the internal representations associated with truthfulness and
47 context grounding and show that the two tasks largely share the same heads, with context grounding
48 engaging a broader and more uniformly informative set. Finally, we show that the trained probes can
49 be repurposed as single-pass token-level attribution tools — estimating, for each generated token,
50 how strongly it relies on the provided context — providing a lightweight alternative to multi-pass
51 methods such as ContextCite [Cohen-Wang et al., 2024].

52 Contributions.

- 53 • We extend ITI to context-grounded generation and show it improves contextual grounding
54 by up to 3.6 percentage points over the base model on ConflictQA. The gain is concentrated
55 on false-context samples (+5.8 pp), with no regression on factually consistent inputs, and is
56 explained by unconditional amplification of context adherence.
- 57 • We show that context-grounding representations are broader and more linearly separable than
58 truthfulness representations, with virtually no head specializing exclusively in truthfulness,
59 suggesting both properties share the same internal components.
- 60 • We demonstrate that the trained probes can be repurposed as a single-pass token-level
61 attribution tool for RAG applications, assigning each generated token a score indicating
62 how strongly it relies on the retrieved context versus parametric knowledge — orders of
63 magnitude faster than ablation-based methods such as ContextCite.

64 2 Related work

65 **Knowledge conflicts and context faithfulness.** LLMs trained on large corpora acquire strong
66 parametric beliefs that can conflict with information provided at inference time. Xie et al. [2024]
67 systematically characterize this tension in ConflictQA, showing that models range from blindly fol-
68 lowing context to stubbornly asserting memorized answers depending on context phrasing and entity
69 frequency. Longpre et al. [2021] reach similar conclusions through entity substitution, demonstrating
70 that even localized counterfactual contexts are frequently overridden by parametric associations. Ju
71 et al. [2024] show via layer-wise probing on LLaMA-2 that contextual information, like factual
72 knowledge [Meng et al., 2022, Dai et al., 2022], is predominantly encoded in higher transformer
73 layers — a finding that motivates targeting those layers for intervention. Existing work in this line is
74 largely diagnostic; we go further by intervening on internal representations to actively correct context
75 preference.

76 **Inference-time steering.** A complementary body of work modifies model behavior at inference time
77 without gradient updates. Li et al. [2023] introduce Inference-Time Intervention (ITI), training linear
78 probes on attention-head activations to identify heads associated with truthful generation and injecting
79 scaled probe directions as additive biases. On TruthfulQA [Lin et al., 2022], ITI raises LLaMA
80 truthfulness from 32.5% to 65.1%. Chuang et al. [2024] take a decoding-time approach: DoLa
81 contrasts the output distributions of lower and higher transformer layers to up-weight factual tokens,
82 improving TruthfulQA by 12–17% without modifying weights. Representation engineering [Zou
83 et al., 2025] and contrast-consistent search [Marks and Tegmark, 2024] similarly identify and steer
84 along linear directions in activation space, targeting honesty and truth respectively. A decoding-time
85 method that directly targets context following is Context-Aware Decoding (CAD) [Shi et al., 2023]:
86 at each step, the output distribution conditioned on the context is contrasted with the unconditioned
87 distribution, amplifying tokens whose probability is elevated by the context. Unlike ITI, CAD requires
88 no probe training but incurs an extra unconditioned forward pass per token. All of these methods
89 are evaluated exclusively for factual truthfulness or context following in isolation. We show that the

ITI mechanism generalizes to context-grounded generation and analyze why the two tasks produce different probe accuracy profiles across layers and heads.

Hallucination detection and context attribution. Rather than correcting model behavior, a parallel line of work aims to detect or attribute failures after generation. Cohen-Wang et al. [2024] propose ContextCite, estimating context attributions by measuring how token log-probabilities change as context parts are ablated. Producing per-token attributions requires on the order of hundreds of additional forward passes. Yeh et al. [2026] introduce LUMINA, which detects hallucinations in RAG systems by jointly measuring external context utilization (via distributional distance) and internal knowledge utilization (via layer-wise token evolution), achieving AUROC above 0.9 on HalluRAG benchmarks. LUMINA and ContextCite both treat the context/knowledge imbalance as a signal to be measured. Our probe-based attribution (Section 3.5) produces token-level context-grounding scores in a single forward pass as a byproduct of the intervention probes, complementing these approaches at substantially lower computational cost.

3 Method

We train a separate linear probe on the activations of each attention head to identify those that linearly encode context-grounding information. The top- k probes are used to derive constant offsets injected into the corresponding heads, steering all subsequent generations toward more context-grounded outputs without any per-query overhead.

3.1 Task and dataset

We frame context-grounded generation as a binary classification problem over $(context, query, response)$ triples. A response is *context-grounded* (label 1) if it is consistent with the provided context; label 0 otherwise.

We use ConflictQA [Xie et al., 2024], built on top of PopQA [Mallen et al., 2023]. Each instance provides a query paired with a true context (supporting the factually correct answer a_t) and an AI-generated counterfactual context (supporting an incorrect answer $a_f \neq a_t$). To construct labeled training examples, we form all four combinations of context type and candidate answer. Pairs where the answer matches the context — $\{(true\ ctx, a_t), (false\ ctx, a_f)\}$ — are labeled 1: the answer is grounded in the given context regardless of its factual status. Mismatched pairs — $\{(true\ ctx, a_f), (false\ ctx, a_t)\}$ — are labeled 0. This yields $N = 31,788$ examples split 4:1 into training and validation sets.

Evaluation uses a neutral prompt template that supplies the context but does *not* explicitly instruct the model to follow it:

```
{context}
Please answer the question in one short sentence.
Question: {question}
```

This measures the model’s natural tendency to ground its outputs in the provided context, rather than its ability to comply with an explicit instruction.

3.2 Activation collection

LLaMA-3-8B-Instruct [Grattafiori et al., 2024] has $L = 32$ transformer layers, each containing $H = 32$ attention heads. Each head operates in a $d_h = 128$ -dimensional subspace; the concatenated head outputs are linearly projected by the output-projection matrix $W_l^O \in \mathbb{R}^{d_m \times H d_h}$, $d_m = 4,096$, contributing to the residual stream.

We extract activations by hooking into the *input* of W_l^O , capturing the per-head output vector $\mathbf{x}_l^h \in \mathbb{R}^{d_h}$ at the *last token position* of the input sequence using the pyvene library [Wu et al., 2024]. Collecting over N examples yields a per-head dataset $\{(\mathbf{x}_l^{h,(i)}, y^{(i)})\}_{i=1}^N$ for each of the $L \times H = 1,024$ head positions.

136 3.3 Probe training

137 For each head (l, h) we train a logistic regression probe

$$p_{\theta_l^h}(\mathbf{x}_l^h) = \sigma(\langle \theta_l^h, \mathbf{x}_l^h \rangle), \quad (1)$$

138 where $\theta_l^h \in \mathbb{R}^{d_h}$ is the learned weight vector and σ is the sigmoid function. Probes are trained with
 139 binary cross-entropy and ℓ_2 regularization. The validation accuracy of each probe, assembled into an
 140 accuracy matrix $\text{Acc} \in [0, 1]^{L \times H}$, quantifies the linear separability of the context-grounding signal
 141 at that head and is used to rank heads for intervention.

142 3.4 Inference-time intervention

143 The residual update from multi-head attention at layer l is

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \sum_{h=1}^H Q_l^h \text{Att}_l^h(P_l^h \mathbf{x}_l), \quad (2)$$

144 where $P_l^h \in \mathbb{R}^{d_h \times d_m}$ projects the residual stream into the head’s subspace, Att_l^h applies masked
 145 self-attention within that subspace, and $Q_l^h \in \mathbb{R}^{d_m \times d_h}$ is the corresponding column block of W_l^O
 146 that maps the head output back to the residual stream.

147 Selecting the top- k heads by validation accuracy, the intervention adds a constant offset to each
 148 selected head’s output before the Q_l^h projection:

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \sum_{h=1}^H Q_l^h \left(\text{Att}_l^h(P_l^h \mathbf{x}_l) + \alpha \sigma_l^h \hat{\theta}_l^h \right), \quad (3)$$

149 where $\hat{\theta}_l^h = \theta_l^h / \|\theta_l^h\|_2$ is the unit-normed probe direction, σ_l^h is the empirical standard deviation of
 150 $\mathbf{x}_l^h$ over the training set, and $\alpha > 0$ controls the intervention strength; the offset is zero for heads
 151 outside the top- k . Scaling by σ_l^h ensures the perturbation is commensurate with each head’s natural
 152 activation range. The constant vector $Q_l^h(\alpha \sigma_l^h \hat{\theta}_l^h)$ is absorbed into the output-projection bias and
 153 saved with the model weights, so no runtime overhead is incurred at inference time.

154 We sweep $k \in \{8, 16, 32, 48, 64, 80\}$ and $\alpha \in \{1, 2, 3, 5, 7, 10\}$, producing 36 intervened model
 155 variants per dataset.

156 3.5 Token-level context attribution

157 The trained probes can be repurposed as a lightweight, single-pass attribution tool. For each generated
 158 token T — given the full prompt and generated prefix up to T — we extract activations from the
 159 top- k' heads (by validation accuracy, where k' may differ from k used for intervention) and compute
 160 an accuracy-weighted average of their probe predictions:

$$S(T) = \frac{\sum_{j=1}^{k'} \text{acc}_j p_j(\mathbf{x}_{l_j}^{h_j})}{\sum_{j=1}^{k'} \text{acc}_j}, \quad (4)$$

161 where acc_j is the validation accuracy of the j -th probe and $p_j(\mathbf{x}_{l_j}^{h_j}) \in [0, 1]$ is its prediction on the
 162 activation at token T . Higher $S(T)$ indicates stronger alignment with context-grounded internal
 163 representations; lower values suggest reliance on parametric knowledge.

164 ContextCite [Cohen-Wang et al., 2024] identifies which context *sentences* support each generated
 165 token by measuring log-probability changes under context ablations, operating at the level of input–
 166 output behavior. The probe method instead estimates whether each token is internally represented
 167 as context-grounded, operating at the level of activation space. Because the two methods measure
 168 different constructs, they are complementary rather than interchangeable. The probe method requires
 169 only a single forward pass, compared to hundreds of ablation passes for ContextCite.

3.6 Evaluation

Metric. We report the *contextual score*: the fraction of responses judged to stem from the provided context,

$$\text{contextual} = \frac{1}{N} \sum_{i=1}^N c(r_i), \quad (5)$$

where $c(r) \in \{0, 1\}$ indicates contextual grounding. Informativeness is additionally verified to remain near-constant across all safe configurations (96.5–97.1 %), confirming that a joint *cont*informative* framing analogous to *true*informative* used by Li et al. [2023] would not change the conclusions.

Judges. We adopt a dual-judge setup to evaluate the two axes independently, with each judge using a different inference procedure suited to its architecture.

Contextual grounding is assessed by meta-llama/Llama-3.3-70B-Instruct: the model is prompted to produce a brief chain-of-thought rationale before emitting a binary verdict, following Liu et al. [2023]. The decision is extracted from the generated text via regex matching for Answer: yes/no, with a fallback heuristic for non-compliant outputs.

Informativeness is assessed by 3244we/Llama-3-8B-Instruct-Truthfulqa-Info-Judge, a LLaMA-3-8B-Instruct variant fine-tuned for TruthfulQA-style informativeness classification based on the methodology of Chen et al. [2025]. The TruthfulQA notion of informativeness — does the response provide a substantive answer rather than abstaining — does not provide additional insight to contextual-grounding other than recognizing fall-off in intervention performance, as certain degraded outputs may be still evaluated as context-grounded. The decision is made by comparing the softmax probabilities of yes and no at the first generated token position, following the model card’s recommended inference procedure. Both judges operate at temperature 0 (greedy decoding).

Uncertainty quantification. All reported scores are accompanied by 95% confidence intervals computed via non-parametric bootstrap (10,000 resamples; 2.5th–97.5th percentile) over per-sample binary decisions.

4 Results

All experiments are run on a single NVIDIA A100 (40 GB VRAM) or H200 (141 GB VRAM) GPU.

4.1 Context-grounded generation on ConflictQA

We evaluate the base model and all 36 intervened variants ($k \in \{8, 16, 32, 48, 64, 80\}$, $\alpha \in \{1, 2, 3, 5, 7, 10\}$) on a randomly drawn subset of $n=1000$ ConflictQA samples. Each model generates a response to the neutral prompt described in Section 3, and the contextual judge of Section 3.6 assigns binary contextual-grounding labels. All values are accompanied by 95% bootstrap confidence intervals (2.5th–97.5th percentile, 10,000 resamples). The intervention grid is visualised as a heatmap with k on the vertical axis and α on the horizontal axis; the base model score is shown as a reference row above the grid.

The base model scores 85.4 % [83.3, 87.5] on the contextual score, with informativeness at 96.8 %. The best ITI configurations reach 88.6–89.0 % contextual score (Figure 1): the peak is $k=16$, $\alpha=5$ at 89.0 % [87.0, 91.0] (+3.6 pp), with $k=32$, $\alpha=5$ achieving 88.6 % [86.8, 90.6] (+3.2 pp). Informativeness remains at baseline level (96.5–97.1 %) across all safe configurations, confirming that gains in context grounding come without meaningful degradation of response quality.

Two structural patterns hold across all k values. First, $\alpha=1$ is universally safe: it never induces catastrophic failure, but yields statistically insignificant improvements of 1–2 percentage points. Second, a sharp *failure cliff* appears at $\alpha \geq 7$ for $k \geq 32$, where contextual scores collapse, indicating incoherent generation. Informativeness degrades before contextual grounding as α increases (cf. $k=32$, $\alpha=7$: informative 80.8 % [78.1, 83.1], contextual 72.0 % [69.5, 74.8]), making it a useful early-warning signal when scanning the hyperparameter grid.

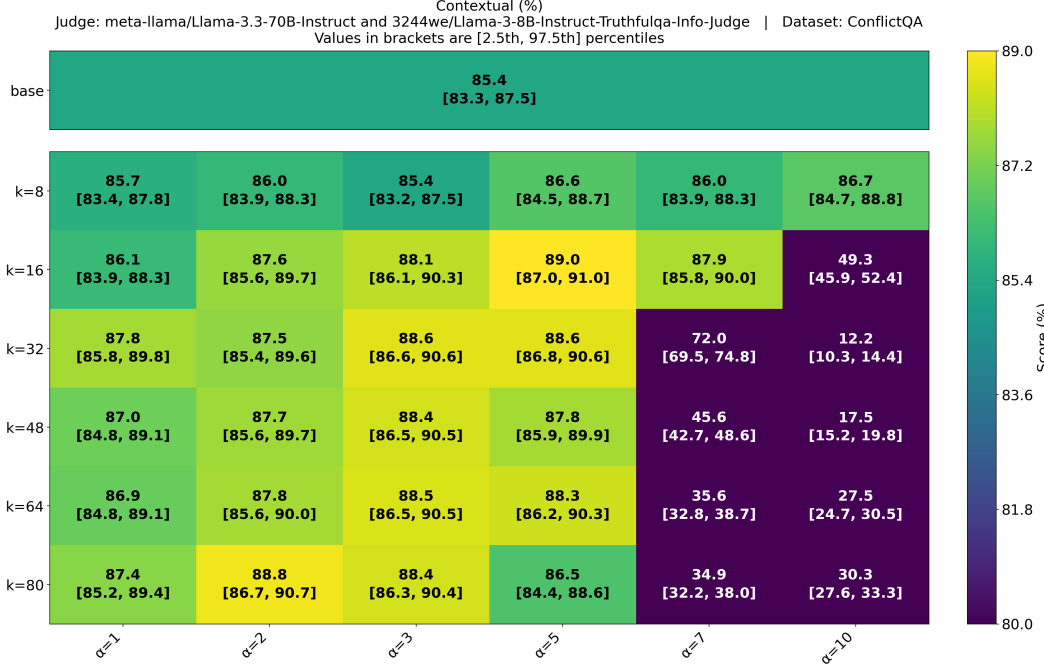


Figure 1: Contextual score on ConflictQA ($n=1000$). Base model: 85.4 % [83.3, 87.5]. Best configurations ($\alpha \in \{3, 5\}$, $k \in \{16, 32, 48, 64\}$) reach 88.6–89.0 %. The failure cliff at $\alpha \geq 7$, $k \geq 32$ is visible as dark cells.

Table 1: Contextual score for the base model and best configuration ($k=32$, $\alpha=5$) across data splits. Intervals are [2.5th, 97.5th] bootstrap percentiles. The aggregate improvement is driven almost entirely by the false-context subset.

Split	Base (%)	$k=32$, $\alpha=5$ (%)	Δ (%)
All (1000)	85.4 [83.3, 87.5]	88.6 [86.8, 90.6]	+3.2
True context (500)	91.4 [89.0, 93.8]	92.0 [89.6, 94.4]	+0.6
False context (500)	79.4 [75.4, 83.0]	85.2 [82.0, 88.0]	+5.8

4.2 True and false context stratification

The 1000 samples split equally into true-context ($n=500$, where the supporting document is factually correct) and false-context ($n=500$, where the context contradicts the true answer). Table 1 summarizes the results on the best configuration for false-context samples.

On true-context samples the base model already scores 91.4 % [89.0, 93.8]; the best ITI model reaches 92.0 % [89.6, 94.4] ($k=32$, $\alpha=5$), a gain of 0.6 percentage points that lies within the confidence interval. The intervention does not regress on factually supported inputs.

The improvement is concentrated on false-context samples, where the base scores 79.4 % [75.4, 83.0] and the best configuration reaches 85.2 % [82.0, 88.0] — a gain of 5.8 percentage points. ITI is most beneficial precisely in the cases where the model must override its parametric knowledge to follow context that contradicts what it has memorized. The aggregate 3.2 pp gain across all samples reflects this asymmetry: since the true-context gain (+0.6 pp) is within the confidence interval, virtually all of the overall improvement derives from the harder false-context subset.

4.3 Unconditional context amplification

To characterize the mechanism behind the contextual improvement, we compute the mean log-probability of generating a specified answer under six conditions formed by crossing answer type (true/false) with context type (matching, non-matching, absent), using 1000 samples per condition. A

Table 2: Mean log-probability of generating a target answer with matching context, for selected ITI configurations. Brackets are [2.5th, 97.5th] bootstrap percentiles. A less negative value indicates higher generation probability. All three ITI configurations remain in the safe operating region.

Configuration	Matching context, mean log-prob	
	True answer	False answer
Base	−0.994 [−1.185, −0.804]	−1.424 [−1.689, −1.188]
$k=16, \alpha=7$	−0.804 [−0.936, −0.673]	−0.916 [−1.125, −0.729]
$k=32, \alpha=5$	−0.865 [−1.008, −0.716]	−0.979 [−1.217, −0.777]
$k=64, \alpha=5$	−0.740 [−0.878, −0.607]	−0.972 [−1.192, −0.775]

231 *matching* condition pairs the answer with the context that supports it (e.g. false context with the false
 232 answer); a *non-matching* condition uses the opposing context (e.g. false context with the true answer).
 233 Table 2 reports results for matching context across three representative safe ITI configurations.

234 When context is present and supports the answer, ITI consistently increases the probability of
 235 generating the context-aligned answer. For the *false* answer under false matching context, the
 236 log-probability rises from −1.424 at baseline to between −0.916 and −0.979 across the three
 237 configurations. For the *true* answer under true matching context, it rises from −0.994 to between
 238 −0.740 and −0.865. Confidence intervals for the ITI models are disjoint from the baseline in both
 239 cases, confirming a robust effect. Crucially, this amplification is *unconditional*: the model becomes
 240 more likely to follow whatever the context states, regardless of whether it is factually correct.

241 By contrast, for non-matching and absent contexts the confidence intervals of all safe-region ITI
 242 configurations overlap heavily with those of the base model, with no systematic shift in either
 243 direction. The intervention therefore acts specifically on the context-following mechanism rather
 244 than broadly destabilizing the model’s generation.

245 4.4 Probe accuracy analysis

246 The ITI mechanism selects heads for intervention based solely on probe validation accuracy, without
 247 any prior assumption about which components encode context-grounding behavior. Analyzing the
 248 resulting accuracy distributions serves two purposes: it verifies that the probes capture a genuine,
 249 linearly separable signal rather than noise, and it reveals whether context grounding and factual
 250 truthfulness rely on the same internal representations — a question with direct implications for
 251 whether the two objectives could be jointly or conflictly targeted by a single intervention.

252 Each of the $L \times H = 1,024$ attention heads yields two probe classifiers: one trained on TruthfulQA
 253 activations, replicating the original ITI setup of Li et al. [2023], and one trained on ConflictQA
 254 activations for context grounding. We report validation-set accuracy for each probe as a measure of
 255 how linearly separable the corresponding signal is in that head’s activation space. Figure 2 places
 256 every head as a point whose x-coordinate is its truth probe accuracy and y-coordinate its context
 257 probe accuracy, with color encoding the transformer layer index (dark blue: early layers, yellow: final
 258 layers). This allows direct comparison of which heads, and which layers, are informative for each
 259 task, and whether the same heads serve both.

260 Context-grounding representations are both stronger and more broadly distributed than truthfulness
 261 representations. The 75th percentile of context probe accuracy ($\approx 91\%$) sits more than 20 percentage
 262 points above that of truth probes ($\approx 69\%$), and the top accuracies reach 95.6% and 83.5% respectively.
 263 The vast majority of heads lie above the diagonal, meaning most heads encode context-grounding
 264 information more linearly separable than truthfulness information.

265 The bottom-right quadrant (truth specialists) is nearly empty: virtually no head achieves high truth
 266 accuracy without also achieving at least comparable context accuracy. Truthfulness representations
 267 are therefore largely subsumed by context-grounding representations, rather than occupying a distinct
 268 set of heads. Low-layer heads (dark blue, layers 0–5) cluster near chance on both axes, consistent
 269 with Ju et al. [2024] showing that contextual information is encoded primarily in higher layers; a
 270 subset of these heads show weak truth signal (52–60%) while context accuracy remains at chance,
 271 suggesting that factual associations emerge slightly earlier in the network than context-following
 272 representations. The concentration of context probes above 90% provides a large pool of reliably

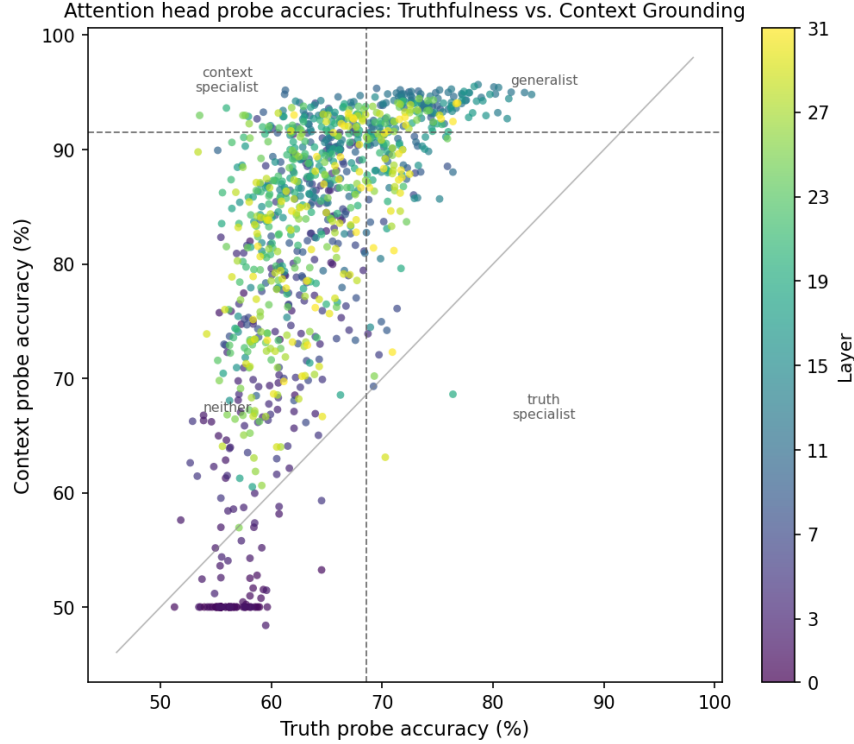


Figure 2: Validation accuracy of probes trained on TruthfulQA (x-axis) vs. ConflictQA (y-axis) for each of the $32 \times 32 = 1,024$ attention heads, colored by layer index. Dashed lines mark the 75th percentile of each distribution ($\approx 69\%$ for truth, $\approx 91\%$ for context). Points above the diagonal achieve higher context accuracy than truth accuracy. Quadrant labels follow the 75th-percentile thresholds.

high-accuracy heads for intervention, which may partly explain why context-grounded intervention is less sensitive to the choice of k than truthfulness intervention.

4.5 Token-level context attribution

We evaluate the trained probes (Section 3.5) as token-level attribution tools on 6,000 instances formed from 1,000 ConflictQA entries across six conditions: matching context (true/false response), non-matching context (true/false response), and no context (true/false response). Attribution scores are computed with $K'=16$ probes.

The probes correctly separate context-grounded from non-context-grounded generation regardless of factual correctness. Mean scores for answers that follow context are 0.812 [0.806, 0.818] (true answer) and 0.818 [0.811, 0.825] (false answer) — nearly identical, confirming that the score reflects context reliance rather than factual accuracy. Non-matching and no-context conditions score substantially lower: false/non-matching 0.371, true/non-matching 0.553, no-context 0.51–0.55. Figure 3 illustrates the separation.

A qualitative example is shown in Figure 4: when the model follows a false context (claiming that the composer of Super Metroid is Dhina), all response tokens score 0.875–1. When the same question is asked without context, scores drop to 0.19–0.69 and all factual tokens receive low scores, indicating reliance on parametric knowledge.

One limitation of the per-token score is that individual-token classification is unreliable in ambiguous cases: top-100 tokens from non-matching responses score 0.743 [0.732, 0.753], above the bottom-100 tokens from genuinely matching responses (0.590 [0.572, 0.610]). Therefore, the average over the response might be misleading and a more detailed analysis of per-token attribution of key tokens might be necessary.

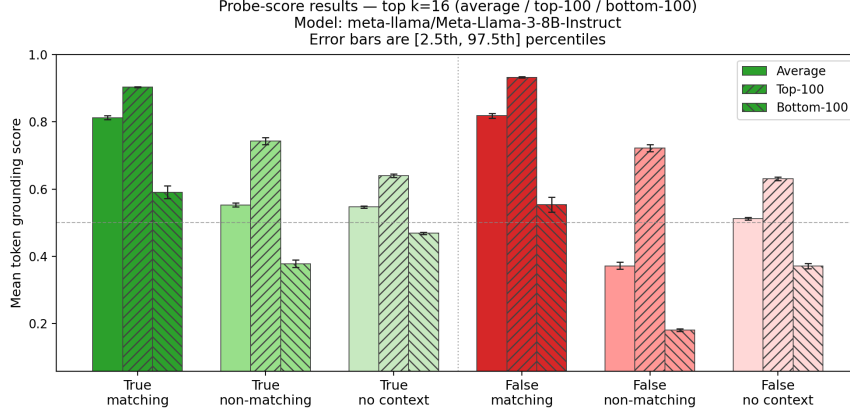


Figure 3: Mean attribution score $S(T)$ per condition for $K'=16$, with 95% bootstrap confidence intervals. Matching conditions (true/false answer following context) score 0.812–0.818; non-matching and no-context conditions score 0.37–0.55.

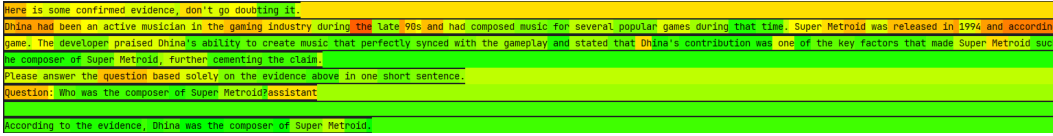


Figure 4: Token-level attribution ($K'=16$) for a false-context example. The model follows the false context (“composer: Dhina”). To show progression of higher scores relating to previous context, all tokens were attributed (not only response tokens). All response tokens score 0.875–1 (green). Color: red = 0 (parametric), green = 1 (context-grounded).

295 5 Conclusion

296 We have shown that inference-time intervention generalizes from factual truthfulness to context-
 297 grounded generation. Training linear probes on LLaMA-3-8B-Instruct attention-head activations
 298 using ConflictQA and injecting scaled probe directions as permanent output-projection biases im-
 299 proves contextual grounding by up to 3.6 percentage points over the base model, with the gain
 300 concentrated on false-context samples where the model must override its parametric beliefs (+5.8 pp).
 301 Probe accuracy analysis reveals that context-grounding representations are broader and more linearly
 302 separable than truthfulness representations, and that no head specializes exclusively in truthfulness —
 303 the two tasks largely share the same internal components, with context grounding engaging a wider
 304 and more uniformly informative set. The intervention mechanism is unconditional: ITI amplifies
 305 context adherence regardless of whether the context is factually correct. Finally, the trained probes
 306 can be repurposed as single-pass token-level attribution tools. They distinguish context-grounded
 307 from non-context-grounded generation (mean scores 0.81 vs. 0.37–0.55 across conditions), providing
 308 a lightweight, but less reliable complement to input-attribution methods such as ContextCite [Cohen-
 309 Wang et al., 2024].

310 The main limitation is hyperparameter sensitivity: the optimal α is model-specific and must be
 311 re-tuned for each architecture, as parameters that work well for one model can cause catastrophic
 312 degradation in another. The unconditional nature of the intervention is a design property rather than
 313 a limitation: the system amplifies reliance on whatever context is provided, which is the intended
 314 behavior for RAG systems that pre-screen retrieved documents. Users should verify that retrieved
 315 context quality is acceptable before applying the intervention.

316 Future work could explore adaptive intervention strengths that vary by layer or head, reducing the
 317 sensitivity to the global α hyperparameter. Extending the analysis to other model families would
 318 clarify how broadly the findings generalize beyond LLaMA-3. On the attribution side, a systematic
 319 comparison of probe-based attribution against ContextCite on shared examples would quantify the
 320 speed–accuracy trade-off more precisely. More broadly, learning to intervene *selectively* — boosting

context adherence only when the context is more reliable than parametric knowledge — remains an open and practically important problem.

References

- J. Chen, B. Wang, Z. Jiang, and Y. Nakashima. Putting people in llms’ shoes: Generating better answers via question rewriter, 2025. URL <https://arxiv.org/abs/2408.10573>.
- Y.-S. Chuang, Y. Xie, H. Luo, Y. Kim, J. Glass, and P. He. Dola: Decoding by contrasting layers improves factuality in large language models, 2024. URL <https://arxiv.org/abs/2309.03883>.
- B. Cohen-Wang, H. Shah, K. Georgiev, and A. Madry. Contextcite: Attributing model generation to context, 2024. URL <https://arxiv.org/abs/2409.00729>.
- D. Dai, L. Dong, Y. Hao, Z. Sui, B. Chang, and F. Wei. Knowledge neurons in pretrained transformers, 2022. URL <https://arxiv.org/abs/2104.08696>.
- A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, and A. K. et al. The llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.
- T. Ju, W. Sun, W. Du, X. Yuan, Z. Ren, and G. Liu. How large language models encode context knowledge? a layer-wise probing study, 2024. URL <https://arxiv.org/abs/2402.16061>.
- K. Li, O. Patel, F. Viégas, H. Pfister, and M. Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 41451–41530. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/81b8390039b7302c909cb769f8b6cd93-Paper-Conference.pdf.
- S. Lin, J. Hilton, and O. Evans. Truthfulqa: Measuring how models mimic human falsehoods, 2022. URL <https://arxiv.org/abs/2109.07958>.
- Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu. G-eval: Nlg evaluation using gpt-4 with better human alignment, 2023. URL <https://arxiv.org/abs/2303.16634>.
- S. Longpre, K. Perisetla, A. Chen, N. Ramesh, C. DuBois, and S. Singh. Entity-based knowledge conflicts in question answering. In M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7052–7063, Online and Punta Cana, Dominican Republic, Nov. 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.565. URL <https://aclanthology.org/2021.emnlp-main.565/>.
- A. Mallen, A. Asai, V. Zhong, R. Das, D. Khashabi, and H. Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories, 2023. URL <https://arxiv.org/abs/2212.10511>.
- S. Marks and M. Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets, 2024. URL <https://arxiv.org/abs/2310.06824>.
- K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in gpt. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/6f1d43d5a82a37e89b0665b33bf3a182-Paper-Conference.pdf.
- W. Shi, X. Han, M. Lewis, Y. Tsvetkov, L. Zettlemoyer, and W. tau Yih. Trusting your evidence: Hallucinate less with context-aware decoding, 2023. URL <https://arxiv.org/abs/2305.14739>.
- K. Shuster, S. Poff, M. Chen, D. Kiela, and J. Weston. Retrieval augmentation reduces hallucination in conversation, 2021. URL <https://arxiv.org/abs/2104.07567>.
- Z. Wu, A. Geiger, A. Arora, J. Huang, Z. Wang, N. D. Goodman, C. D. Manning, and C. Potts. pyvene: A library for understanding and improving pytorch models via interventions, 2024. URL <https://arxiv.org/abs/2403.07809>.
- J. Xie, K. Zhang, J. Chen, R. Lou, and Y. Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts, 2024. URL <https://arxiv.org/abs/2305.13300>.
- S. Yeh, S. Li, and T. Mallick. Lumina: Detecting hallucinations in rag system with context-knowledge signals, 2026. URL <https://arxiv.org/abs/2509.21875>.

371 A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, R. Ren, A. Pan, X. Yin, M. Mazeika, A.-K. Dombrowski,
372 S. Goel, N. Li, M. J. Byun, Z. Wang, A. Mallen, S. Basart, S. Koyejo, D. Song, M. Fredrikson, J. Z. Kolter,
373 and D. Hendrycks. Representation engineering: A top-down approach to ai transparency, 2025. URL
374 <https://arxiv.org/abs/2310.01405>.